

Lesson 22: System of Three Linear Equations and Matrix

Criteria	Exceptional (4)	Proficient (3)	Developing (2)	Beginning (1)
Augmented Matrix Setup	Correctly translates all 3 equations into a 3x4 matrix; signs and constants are perfectly placed.	1–2 minor sign errors or placement errors in the matrix.	Major errors in setting up the matrix, but structure is somewhat recognizable.	Incorrect matrix structure (e.g., mixing coefficients and variables).
Row Operations Application	Consistently applies elementary row operations accurately. Work is organized, logical, and easy to follow.	Most row operations are accurate; 1–2 minor arithmetic errors that do not break the method.	Multiple errors in arithmetic or invalid row operations used (e.g., replacing row without scaling).	Steps are illogical or do not follow Gauss-Jordan elimination principles.

RREF Target Achieved	Successfully transforms the matrix to $[I B]$ with leading 1s on the diagonal and 0s elsewhere.	Reaches Row Echelon Form (REF) correctly, or has 1–2 errors in finalizing the RREF.	Matrix is not in RREF; fails to create 0s above or below leading 1s.	Matrix remains mostly unreduced.
Interpretation of Solution	Accurately identifies unique (x, y, z) solution, or correctly identifies no/infinite solutions based on RREF.	Correct interpretation of solution but with minor arithmetic errors from previous steps.	Incorrectly interprets RREF (e.g., misreads a row of zeros).	Cannot identify the solution from the final matrix.
Verification & Documentation	Final solution is verified by plugging back into original equations. All work shown.	Solution is verified, but minor errors exist in the verification arithmetic.	Verification attempt is made but not completed or incorrectly performed.	No verification performed.